across disconnected systems creates dangerous delays when equipment safety demands immediate decisions.

Each step—matching the sensor to its specifications, linking it to system diagrams, connecting it to maintenance records—contextualizes raw data by adding layers of meaning. Yet in most facilities, this happens manually. The effort required to extract, match, and interpret data across siloed systems is expensive and error-prone. Worse, this critical knowledge often exists only as tacit expertise held by senior staff, making it vulnerable to loss and impossible to scale.

Digital twins solve this by encoding context directly into their model of the physical world. By capturing relationships between sensor readings and their associated equipment, specifications, maintenance history, and interconnected systems, a digital twin automatically transforms a measurement like 85° into the right interpretation, whether that’s an urgent alert or routine confirmation of normal operation. The contextual framework that experts apply intuitively becomes explicit, queryable, and consistently available.

5.2.1 Types of context

Context transforms raw data into actionable information. Digital twins require four types of context to build a complete understanding of physical systems:

- *Spatial context* is the location of an entity. It links data to physical locations, whether absolute geographical coordinates or relative positions between objects. Knowing that a sensor is 2 meters from a west-facing window (in the southern hemisphere) explains why it shows temperature spikes in the afternoon. Spatial context enables meaningful visualization and reveals physical relationships that affect system behavior.
- *Temporal context* connects each data point to a specific moment in time. A temperature reading of 20°C is just a number; knowing it was recorded at 3:00 PM, rose from 18°C at 2:00 PM, and typically peaks at 22°C by 4:00 PM reveals patterns, trends, and potential anomalies.
- *Relational context* captures how things relate to each other. It answers questions like "which valve controls this pipe?"" or "what maintenance schedule applies to this pump?" These relationships link data points into a web of meaning that enables holistic reasoning.